

SETS IN WHICH EQUAL PRODUCTS OF DISTINCT ELEMENTS FORCE EQUAL CARDINALITIES: A NEGATIVE ANSWER TO ERDŐS PROBLEM #786

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ABSTRACT. Call a set A of positive integers *admissible* if, whenever $a_1 \cdots a_r = b_1 \cdots b_s$ with $a_1, \dots, a_r$ distinct elements of A and $b_1, \dots, b_s$ distinct elements of A , necessarily $r = s$. Erdős asked whether admissible sets can have density $1 - \varepsilon$ for every $\varepsilon > 0$, and whether $\{1, \dots, N\}$ always contains an admissible subset of size $(1 - o(1))N$. For the variant in which repetitions are allowed both questions were answered negatively by Erdős, Ruzsa and Sárközy and by Granville and Soundararajan; for products of distinct elements, question (i) was answered only recently (with density bound $7/8$) and question (ii) has remained open. We show that every admissible $A \subseteq \{1, \dots, N\}$ satisfies $\sum_{a \in A} 1/a \leq \frac{1}{2} \log N + (\log \log N + 2)^2$, and that there is an absolute constant $\eta > 0$ such that every admissible $A \subseteq \{1, \dots, N\}$ has $|A| < (1 - \eta)N$ for all large N . Both questions therefore have negative answers. The proofs are elementary; the second rests on a coupling that replaces the largest divisor of an integer composed of small primes, which avoids the divisor-function losses inherent in counting quotients along a multiplication table. Both negative answers are formally verified in Lean 4 against the statements of the Formal Conjectures project.

1. INTRODUCTION

For a set A of positive integers consider the property

$$(P) \quad S, T \subseteq A \text{ finite}, \quad \prod_{s \in S} s = \prod_{t \in T} t \implies |S| = |T|.$$

Equivalently, an identity $a_1 \cdots a_r = b_1 \cdots b_s$ between *distinct* elements a_i of A and distinct elements b_j of A forces $r = s$ (common elements of the two sides may be cancelled). We call such sets *admissible*. Note that $1 \notin A$, and that every subset of an admissible set is admissible. The integers $\equiv 2 \pmod{4}$ form an admissible set of density $1/4$, since the 2-adic valuation counts the factors on each side.

Erdős [2, 3] asked the following (Problem #786 of [1]).

- (i) Is there, for every $\varepsilon > 0$, an admissible set of density greater than $1 - \varepsilon$?
- (ii) Is there always an admissible $A \subseteq \{1, \dots, N\}$ with $|A| \geq (1 - o(1))N$?

Selfridge observed that the integers having exactly one prime factor, counted with multiplicity, from a suitable block of consecutive large primes (with reciprocal sum just below 1) form an admissible set of density $1/e - \varepsilon$, and Tao observed (see [1]) that the integers $n \leq N$ with exactly one prime factor, counted with multiplicity, exceeding $N^{1/(1+\sqrt{e})}$ form an admissible set of size $(1 - c + o(1))N$ with $c = 0.1715 \dots$

If one allows repetitions (so that (P) is required for multisets), then a finite set is admissible precisely when there is a completely additive $F : \mathbb{N} \rightarrow \mathbb{Q}$ with $F \equiv 1$ on A (by linear duality), and both questions are settled: Erdős, Ruzsa and Sárközy [4] obtained the density bound $1/2$, and Tao noted that a theorem of Granville and Soundararajan [8] gives $|A| \leq (1 - c + o(1))N$ with the constant c above, which is sharp. The formulation in [3, p. 114], however, explicitly concerns

distinct elements; there Erdős states that Ruzsa had shown that both answers are negative, and that an infinite admissible set has upper density $< 1/e$. No proof has appeared, and the problem page [1] remarks that Erdős may have had the repetition-allowed results in mind; the distinct version is listed there as open. Recently Gessel [7] posted a proof, accompanied by a Lean formalization, that an admissible set with natural density δ has $\delta \leq 7/8$, which answers (i); question (ii) remained open. The repetition-free condition is weaker, and there is no additive certificate: for instance $\{2, 3, 4\}$ is admissible but $2 \cdot 2 = 4$.

Theorem 1. *For every $N \geq 286$ and every admissible $A \subseteq \{1, \dots, N\}$,*

$$\sum_{a \in A} \frac{1}{a} \leq \frac{1}{2} \log N + (\log \log N + 2)^2.$$

Consequently an admissible set has upper logarithmic density at most $1/2$, and natural density at most $1/2$ whenever the latter exists.

Theorem 2. *There is an absolute constant $\eta > 0$ such that for all sufficiently large N , every admissible $A \subseteq \{1, \dots, N\}$ satisfies $|A| < (1 - \eta)N$. One may take $\eta = e^{-5000}$ for $N \geq e^{200000}$.*

Theorem 2 answers (ii) negatively, and also (i): an admissible set of density $> 1 - \eta/2$ would contain admissible sets $A \cap [1, N]$ of size $\geq (1 - \eta)N$. Theorem 1 gives the stronger bound $1/2$ for (i), matching the repetition-allowed bound of [4]. We make no attempt to optimize η ; it seems natural to conjecture, following Tao, that the optimal deficit is $0.1715\dots$, i.e. that the largest admissible subsets of $\{1, \dots, N\}$ have size $(1 - 0.1715\dots + o(1))N$, in the distinct setting as well. Some mild numerical evidence points the same way: an exact computation for $N \leq 60$ (two independent programs, available with [10]) shows that for $34 \leq N \leq 60$ the largest admissible subset of $\{1, \dots, N\}$ has the same size as in the repetition-allowed version, and for $39 \leq N \leq 60$ an extremal set is $\{n \leq N : n \text{ has exactly one prime factor } \geq 5, \text{ counted with multiplicity}\}$, a level set of a completely additive function.

Formal verification. The negative answers to both questions, in the exact form stated in the Formal Conjectures project [5] (`erdos_786.parts.i` and `erdos_786.parts.ii`, with the answer resolved to `False`), are proved in Lean 4 with Mathlib; the only analytic input beyond Mathlib is the explicit form of Mertens' theorems, taken (vendored) from the `PrimeNumberTheoremAnd` project [9]. The formal proof follows Section 4 with unspecified constants. See Section 5.

Notation. $H(S) = \sum_{s \in S} 1/s$, $H_j = \sum_{n \leq j} 1/n$; $v_p(n)$ is the exponent of p in n ; $\tau(n)$ the number of divisors; p, q denote primes. For a positive rational $q \neq 1$, a q -pair in A is a pair (x, qx) with $x, qx \in A$, and a q -matching is a family of q -pairs whose endpoints are pairwise distinct.

2. TWO COMBINATORIAL LEMMAS

Lemma 3 (Lifting). *Let $a \in A$ and $a = q_1 \cdots q_t$ with positive rationals $q_j \neq 1$ (repetitions allowed). If for each j the q_j -pairs in A contain a matching of size at least $2t$, then A is not admissible.*

Proof. Choose (x_j, y_j) , $y_j = q_j x_j$, for $j = 1, \dots, t$ in turn from the matching of q_j , avoiding a and all endpoints chosen earlier. At step j there are $2j - 1$ forbidden integers, each lying in at most one pair of a matching, so a pair is available. Then $a, x_1, y_1, \dots, x_t, y_t$ are distinct elements of A and $a \prod x_j = \prod y_j$, with $t + 1 \neq t$ factors. (If $t = 0$ then $a = 1$, which is excluded by (P).) $\square$

Lemma 4. *Let $q > 1$. Any R distinct q -pairs contain a q -matching of size at least $R/2$.*

Proof. The graph with edges $\{x, qx\}$ has maximum degree 2 and is acyclic (values increase along $x \mapsto qx$), so it is a union of paths; take alternate edges. $\square$

A first consequence: if $R_p = \#\{x : x, px \in A\}$ and $a = p_1 \cdots p_k \in A$ with every $R_{p_i} \geq 4 \log_2 N$, then greedily choosing fresh pairs $(x_i, p_i x_i)$ (at step i at most $4i - 2$ candidates are blocked) gives $a \prod x_i = \prod p_i x_i$. Since $R_p \geq \lfloor N/p \rfloor - 2(N - |A|)$, this already forces all y -smooth integers with $y \approx N/(2(N - |A|))$ out of A ; but that yields only a deficit $N \exp(-(1 + o(1))\sqrt{\log N \log \log N})$.

3. LOGARITHMIC DENSITY: PROOF OF THEOREM 1

Lemma 5 (Packing). *Let P be a finite set of primes and $S \subseteq \{1, \dots, N\}$ such that every element of S is divisible by some $p \in P$, and x, px never both lie in S for $p \in P$. Then $H(S) \leq \frac{1}{2}H_N + \frac{1}{2}$.*

Proof. Fix m and write $m = b \prod_{i \leq r} p_i^{e_i}$ with $p_1, \dots, p_r$ the primes of P dividing m and $(b, \prod p_i) = 1$. For each $c \mid b$ the divisors $c \prod p_i^{f_i}$, $0 \leq f_i \leq e_i$, form a box $B = \prod \{0, \dots, e_i\}$; the exponent vectors of members of S avoid 0 and contain no two vectors differing by 1 in one coordinate. Pair B into adjacent pairs along a coordinate with e_i odd if there is one; otherwise pair $B \setminus \{0\}$ along the first nonzero coordinate via $(1, 2), (3, 4), \dots$. Hence $\#\{d \mid m : d \in S\} \leq \tau(m)/2$. Summing over $m \leq N$ gives $\sum_{s \in S} \lfloor N/s \rfloor \leq \frac{1}{2} \sum_{d \leq N} \lfloor N/d \rfloor$; writing $r_d = N/d - \lfloor N/d \rfloor \in [0, 1)$,

$$NH(S) \leq \frac{N}{2}H_N + \frac{1}{2} \left(\sum_{s \in S} r_s - \sum_{d \notin S} r_d \right) \leq \frac{N}{2}H_N + \frac{N}{2}. \quad \square$$

Proof of Theorem 1. Let $h = \lfloor \log_2 N \rfloor$ and call p *bad* if $R_p < 4h$; let P be the set of bad primes. Every $a \in A$ has a bad prime factor, by the consequence of Lemma 3 noted above (for $a = p_1 \cdots p_k$ with no bad factor, $k \leq h$ and fresh pairs exist). Let $E = \{px : p \in P, x, px \in A\}$ and $S = A \setminus E$. Then S satisfies the hypotheses of Lemma 5, so $H(S) \leq \frac{1}{2}H_N + \frac{1}{2}$, while

$$H(E) \leq \sum_{p \in P} \frac{1}{p} \sum_{x: x, px \in A} \frac{1}{x} \leq H_{4h-1} \sum_{p \leq N} \frac{1}{p}.$$

By Rosser-Schoenfeld [11, Thm. 5], $\sum_{p \leq N} 1/p \leq \log \log N + 1$ for $N \geq 286$; also $H_{4h-1} \leq 1 + \log(4 \log N / \log 2) < \log \log N + 3$ and $\frac{1}{2}H_N + \frac{1}{2} \leq \frac{1}{2} \log N + 1$. With $\ell = \log \log N$ we get $H(A) \leq \frac{1}{2} \log N + 1 + (\ell + 3)(\ell + 1) = \frac{1}{2} \log N + (\ell + 2)^2$. The statements about densities follow by applying this to $A \cap [1, X]$ and, for natural density, by partial summation. $\square$

Remark 6. Harmonic mass is insensitive to $[N/K, N]$ (the interval $[N/\log N, N]$ has $(1 - o(1))N$ elements but harmonic mass $\approx \log \log N$), so Theorem 1 does not bear on question (ii).

4. PROOF OF THEOREM 2

4.1. Why a new idea is needed. A natural approach to (ii) is to connect a large prime p to 1 through ratios v/u close to 1 between integers $u, v \leq \sqrt{N}$, and to charge each missing ratio edge to the missing integers mt or nt occurring in the “trials” (mt, nt) , $m \in (K, 2K]$. Carried out in dyadic bands, this charges each failed trial to a divisor of a missing integer, so missing integers are weighted by their number of divisors; one obtains only a deficit bound with a polylogarithmic loss, $N - |A| \gg N/(\log N)^C$. The loss is intrinsic to this kind of charging. By Ford’s theorem [6, Thm. 1], uniformly for $3 \leq K \leq \sqrt{N}$ the number of $n \leq N$ having a divisor in $(K, 2K]$ is $\asymp N(\log K)^{-\delta_F}(\log \log K)^{-3/2}$ with $\delta_F = 1 - \frac{1 + \log \log 2}{\log 2}$, which is $o(N)$ as $K \rightarrow \infty$; yet deleting these integers kills every trial (mt, nt) with $m \in (K, 2K]$. The proof below avoids this by pairing pz with a partner obtained from z through a *canonical* factorization, so that every integer is hit with essentially uniform probability.

4.2. Set-up. Fix $N \geq N_1 = \lceil e^{200000} \rceil$ and $A \subseteq \{1, \dots, N\}$; let $E = \{1, \dots, N\} \setminus A$, $D = |E|$, $\delta = D/N$, and suppose $\delta \leq \eta = e^{-5000}$. Put

$$P = N^{1/32}, \quad M = \lceil 4(\log N)^2 \rceil, \quad c_0 = e^{-1400}, \quad p_0 = e^{3200}.$$

A prime $p \leq N$ is *heavy* if $|E \cap p\mathbb{N}| \geq N/(16p)$; let $\mathcal{H}$ be the set of heavy primes and $S = \sum_{p \in \mathcal{H}} 1/p$. A rational $q \neq 1$ is *rich* if the q -pairs in A contain a matching of size M ; q is rich iff q^{-1} is. For a set Q of primes, $k_Q(z)$ denotes the largest divisor of z all of whose prime factors lie in Q .

4.3. Step 1: heavy primes.

Lemma 7. $S \leq 3072\delta$. In particular $S + \eta < c_0/2$.

Proof. Let $\omega(n) = \#\{p \in \mathcal{H} : p \mid n\}$. Averaging over $n \leq N$, $\mathbb{E}\omega \geq S - |\mathcal{H}|/N$ and $\mathbb{E}\omega^2 \leq S + S^2$, so $\mathbb{E}(\omega - S)^2 \leq S + 2S|\mathcal{H}|/N \leq 3S$. By Cauchy-Schwarz $\sum_{n \in E} \omega(n) \leq DS + \sqrt{3DNS}$, whereas heaviness gives $\sum_{n \in E} \omega(n) = \sum_{p \in \mathcal{H}} |E \cap p\mathbb{N}| \geq NS/16$. Hence $(\frac{1}{16} - \delta)S \leq \sqrt{3\delta S}$, i.e. $S \leq 3\delta/(\frac{1}{16} - \delta)^2 \leq 3072\delta$. $\square$

4.4. Step 2: smooth numbers.

Lemma 8. If $y \geq e^{200}$ and $y^{16} \leq X \leq y^{56}$, then at least c_0X integers in $[X/2, X]$ are y -smooth.

Proof. Let $u = \log X / \log y$, $a = (u - 1)/112$, $b = (u - \frac{1}{2})/112$, so $\frac{1}{8} < a < b < \frac{1}{2}$ and $b/a \geq \frac{111}{110}$. By [11, Thm. 5], $|\sum_{p \leq z} 1/p - \log \log z - B_1| \leq 1/\log^2 z$ for $z \geq 286$, whence the primes $\mathcal{I}$ in $(y^a, y^b]$ satisfy $\sum_{q \in \mathcal{I}} 1/q \geq \log \frac{111}{110} - \frac{128}{200^2} > \frac{1}{200}$. For an ordered 112-tuple from $\mathcal{I}$ with product Q we have $X/y < Q \leq X/\sqrt{y}$, and the $\geq X/(3Q)$ integers $r \in [X/(2Q), X/Q]$ give y -smooth $n = rQ \in [X/2, X]$. There are at least $\frac{X}{3} (\sum_{q \in \mathcal{I}} 1/q)^{112}$ such representations, and each $n \leq y^{56}$ has at most 448 prime factors exceeding $y^{1/8}$, hence at most 448^{112} representations. So there are at least $X/(3 \cdot 89600^{112}) > e^{-1400}X$ such n . $\square$

4.5. Step 3: the Q -part is usually small.

Lemma 9. Let $y \geq e^{200}$, Q a set of primes $\leq y$, and z uniform in $[1, T]$. Then $\mathbb{P}(k_Q(z) > y^{40}) \leq \frac{1}{10}$.

Proof. Chebyshev's bound $\vartheta(t) < 3t$ (from $\prod_{n < p \leq 2n} p \mid \binom{2n}{n}$) and partial summation give $\sum_{q \leq y} \frac{\log q}{q-1} \leq 3 \log y + 7 \leq 4 \log y$. Hence $\mathbb{E} \log k_Q(z) = \sum_{q \in Q} \sum_{j \geq 1} (\log q) [T/q^j]/T \leq 4 \log y$, and Markov's inequality gives the claim. $\square$

4.6. Step 4: small primes. Every prime $p \leq p_0$ is rich: there are at least $\lfloor N/p \rfloor - 2D \geq N/(2p_0) - p$ -pairs, hence a matching of size $N/(4p_0) \geq M$ by Lemma 4. We also use $N^{57/64}/8 \geq M$ and $P \geq p_0$, valid for $\log N \geq 200000$.

4.7. Step 5: replacing the largest smooth part.

Lemma 10. Let p be a non-heavy prime with $p_0 < p \leq P$, $y = p^{1/16}$, and Q the set of non-heavy primes $\leq y$. There are Q -smooth integers k, m with $k \leq y^{40}$ and $pk/2 \leq m < pk$ such that pk/m is rich.

Proof. For Q -smooth $k \leq y^{40}$ let $\mathcal{M}_k$ be the set of Q -smooth $m \in [pk/2, pk]$. Since $y^{16} \leq pk \leq y^{56}$, Lemma 8 gives c_0pk y -smooth integers in $[pk/2, pk]$, of which at most pkS are divisible by a heavy prime; by Lemma 7, $|\mathcal{M}_k| \geq \gamma pk$ with $\gamma = c_0/2$. As $p \notin Q$, $pk \notin \mathcal{M}_k$.

Let $T = \lfloor N/p \rfloor$ and let z be uniform in $[1, T]$; write $z = ku$ with $k = k_Q(z)$. If $k \leq y^{40}$, pick $m \in \mathcal{M}_k$ uniformly and form the pair $(mu, pku) = (mu, pz)$, both $\leq N$. Failures: (a) $k > y^{40}$,

probability $\leq \frac{1}{10}$ by Lemma 9; (b) $pz \in E$, probability $< \frac{N/(16p)}{T} \leq \frac{1}{8}$ as p is not heavy; (c) $mu \in E$. Given $b = mu$, m is the largest Q -smooth divisor of b (as u has no prime factor in Q), so m and u are determined by b , and $z = ku$ with $m/p \leq k \leq 2m/p$. Hence

$$\mathbb{P}(mu = b) \leq \frac{1}{T\gamma p} \sum_{m/p \leq k \leq 2m/p} \frac{1}{k} \leq \frac{2}{T\gamma p} \leq \frac{4}{\gamma N},$$

and $\mathbb{P}(mu \in E) \leq 4\delta/\gamma \leq 8e^{-3600} < \frac{1}{8}$. So with probability $> \frac{1}{2}$ both mu and pz lie in A . For Q -smooth $k \leq y^{40}$ and $m \in \mathcal{M}_k$ let $s_{k,m}$ be the number of u with $ku \leq T$, $k_Q(ku) = k$, $mu \in A$ and $pku \in A$. The success probability equals $\frac{1}{T} \sum_k \frac{1}{|\mathcal{M}_k|} \sum_{m \in \mathcal{M}_k} s_{k,m} \leq \frac{1}{T} \sum_k \max_m s_{k,m}$, so $\sum_k \max_m s_{k,m} > T/2$; as there are at most y^{40} values of k , some (k, m) has $s_{k,m} \geq T/(2y^{40}) \geq N/(4p^{7/2}) \geq N^{57/64}/4$; the corresponding (mu, pk_u) are distinct (pk/m) -pairs, and Lemma 4 shows that pk/m is rich. $\square$

The uniqueness of the decomposition $z = k_Q(z) \cdot u$ is what removes the divisor-function weight: with an arbitrary divisor in place of $k_Q(z)$, a given b would be reached through each of its divisors.

4.8. Step 6: short words.

Lemma 11. *Every non-heavy prime $p \leq P$ is a product of at most $16(\log p)^2$ rich rationals and inverses of rich rationals.*

Proof. Induct on p . For $p \leq p_0$ use Step 4. Otherwise take k, m from Lemma 10; their prime factors are non-heavy and $\leq p^{1/16} < p$. Concatenating the words of the prime factors, a Q -smooth v has a word of length at most $16 \sum_q v_q(v)(\log q)^2 \leq (\log p) \log v$. Writing $p = \frac{pk}{m} \cdot m \cdot k^{-1}$ and using $k \leq p^{5/2}$, $m \leq pk$, the length is at most $1 + (\log p)(\log m + \log k) \leq 1 + 6(\log p)^2 \leq 16(\log p)^2$. $\square$

4.9. Step 7: conclusion. Apply Lemma 8 with $y = P$, $X = N$: at least $c_0 N$ integers in $[N/2, N]$ are P -smooth; at most NS are divisible by a heavy prime and at most D lie in E , so some $a \in A \cap [N/2, N]$ has all its prime factors non-heavy and $\leq P$. By Lemma 11, $a = q_1 \cdots q_t$ with each q_j rich (all rich ratios are $\neq 1$ by definition) and $t \leq 16(\log P) \log a \leq \frac{1}{2}(\log N)^2$, so $M \geq 2t$, and Lemma 3 shows that A is not admissible. This proves Theorem 2. $\square$

5. FORMALIZATION

The Lean 4 development [10] (Lean v4.33.1, Mathlib commit 0df444a) depends only on Mathlib; the explicit Mertens estimates $\sum_{p \leq x} \frac{\log p}{p} = \log x + O(1)$ and $\sum_{p \leq x} \frac{1}{p} = \log \log x + M + O(1/\log x)$, with explicit error terms, are vendored from PrimeNumberTheoremAnd [9] and replace [11]. It proves:

- **Erdos786.main:** there is $\eta > 0$ such that for all large N every $A \subseteq \{1, \dots, N\}$ with $|A| \geq (1 - \eta)N$ contains distinct $a_1, \dots, a_r, b_1, \dots, b_s$ with $r \neq s$ and $\prod a_i = \prod b_j$. The proof follows Section 4 with unspecified constants: Lemma 3 and a greedy version of Lemma 4 (matching of size $\geq R/3$), Step 1 (`heavy_recip_sum_le`), Step 2 (`smooth_count`), Step 3 (`qPart_tail`), Step 5 in counting form (`core_pairs`), Step 6 (`short_word`) and the assembly (`main_of`).
- **erdos_786.parts.ii** and **erdos_786.parts.i** of [5], in the form `False` $\leftrightarrow$ (statement), where the statements and all definitions they use (`Set.IsMulCardSet`, `Set.HasDensity`, `Set.partialDensity`) are copied verbatim. Part (i) is deduced from `main` as explained after Theorem 2.

`#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. Theorem 1 is not formalized.

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